

Week 9 Tutorial

Relative valuation using market comparables

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Agenda

- 8-5
- 8-9
- 8-12

Review 1: Relative valuation

- This valuation using market comparables (*or comps*) involves a four-step process.
 1. Identify similar or comparable investments and recent market prices for each **(most critical step)**
 2. Calculate a valuation metric for use in valuing the asset. (e.g., *Enterprise value / EBITDA*; *P/E* ratio)
 3. Calculate an initial estimate of value
 4. Refine or tailor your initial valuation estimate to the specific characteristics of the investment

Review 1: Relative valuation – Helix Corp

1. Enterprise valuation using **EBITDA Multiples** (textbook, P272)

$$\begin{aligned}\text{Owner's Equity} &= \text{Enterprise Value} - (\text{Interest-Bearing Debt} - \text{Cash}) \quad (8.5a) \\ &= \text{Enterprise Value} - \text{Net Debt}\end{aligned}$$

You may recall from our discussion of the firm's cost of capital in Chapter 4 that the enterprise value of a firm is defined as the sum of the values of the firm's interest-bearing debt and its equity minus the firm's cash balance on the date of the valuation. For example, on March 18, 2014, Airgas Inc. (ARG)⁵ had an enterprise value of \$10.454 billion, which can be computed as follows:

Elements of the Firm's Balance Sheet	Values (000)
Non-interest-bearing debt	\$ 2,335
Interest-bearing debt (short and long term)	2,607.82
Common equity (market value = price per share × shares outstanding)	7,930.00
Firm value	<u>\$ 12,873</u>
Less: non-interest-bearing debt	(2,335.14)
Less: cash and equivalents	(83.39)
Equals: enterprise value	<u><u>\$ 10,454</u></u>

Review 1: Relative valuation – Helix Corp

The Airgas EBITDA Multiple

To calculate the Airgas EBITDA multiple, we solve the following equation:

$$\text{Enterprise Value}_{\text{Airgas 2014}} = \text{EBITDA}_{\text{Airgas 2014}} \times \frac{\text{EBITDA}}{\text{Multiple}} \quad (8.6)$$

For example, on March 18, 2014, Airgas' EBITDA was \$943 million and its enterprise value was \$10,454 billion; this results in an EBITDA multiple for Airgas of 11.09. In the next section, we look at how the Airgas EBITDA multiple, along with multiples of similar firms, can be used to value a privately held company.

Table 8-2 Comparable-Firm EBITDA Multiples for Helix Corporation (\$ Millions)

(\$ millions)	Air Products (APD)	Praxair, Inc. (PRAX)	Applied Industrial Technologies, Inc (AIT)	AirGas (ARG)	Average
Enterprise Value	31,140	47,400	2,010	10,454	
EBITDA	2,481	3,703	197	943	
Multiples	12.55	12.80	10.20	11.09	11.66

Review 1: Relative valuation – Helix Corp

Example: Valuing a Privately Held Firm

To explore the use of an EBITDA multiple, we consider the valuation of Helix Corporation, a privately owned company operating out of Phoenix, Arizona, that is a potential acquisition candidate. Because it is privately owned, a potential acquirer cannot observe its market value. However, the acquirer can use similar firms that *are* publicly traded to infer a value for Helix by using the appropriate EBITDA valuation ratio.

EBITDA		\$ 10,000,000
Enterprise value = $11.66 \times \$10,000,000$		116,600,000
Plus: cash		2,400,000
Less: interest-bearing debt		(21,000,000)
Equity value		\$ 98,000,000

Review 1: Relative valuation - ExxonMobil

One approach would be to use EBITDA multiples to estimate enterprise value, as described in the last section, and then subtract the firm's interest-bearing debt to determine the value of its equity. While this approach is used in practice, it is not the main approach used when valuing a firm's equity. Instead, equity analysts tend to focus their attention on estimating the earnings of the firms they evaluate and then use the **price-to-earnings (P/E) ratio** to evaluate the price of the common stock.

The price-to-earnings (P/E) multiple valuation approach is defined in Equation 8.8:

$$\left(\frac{\text{Price per Share for a Comparable Firm}}{\text{Earnings per Share for a Comparable Firm}} \right) \times \text{Earnings per Share for the Firm Being Valued} = \text{Estimated Value of the Firm's Equity} \quad (8.8)$$

P/E multiple

As we just mentioned, the P/E valuation approach is used to estimate the value of the *firm's equity*. It is not used to find the enterprise value, as we did using the EBITDA valuation method.

Review 1: Relative valuation - ExxonMobil

2. Equity valuation using **Price-to-Earnings Multiples** (textbook, P279)

The chemical division of ExxonMobil is one of the largest chemical companies in the world.⁸ The chemical division earned \$3.428 billion in 2004, compared to \$1.432 billion in 2003 and \$830 million in 2002. If the company were to consider selling the division via an IPO, a key question it would want to answer would be, “How much can we expect to receive for the sale of the equity?” We can get a quick glimpse at the magnitude of this equity value by applying an average P/E ratio of similar firms to the chemical division’s earnings. Table 8-3 contains P/E ratios for seven of the largest chemical companies in the world. They have an average P/E multiple of 14.28, which implies that, as a first approximation, ExxonMobil’s chemical division is likely to be worth about \$48.94 billion = $14.28 \times \$3.428$ billion.

Review 1: Relative valuation - ExxonMobil

Table 8-3 Chemical Company Price/Earnings Ratios (August 16, 2005)

	Share Price	÷	EPS	=	P/E Ratio
BASF AG (BF)	\$ 70.47		\$ 5.243		13.44
Bayer AG (BAY)	35.64		1.511		23.59
Dow Chemical (DOW)	47.40		4.401		10.77
E I DuPont (DD)	41.00		2.572		15.94
Eastman Chemical (EMN)	51.69		5.75		8.99
FMC (FMC)	59.52		5.729		10.39
Rohm & Hass (ROH)	45.02		2.678		16.81
Average					14.28

Review 1: Relative valuation – trailing (forward) P/E

The price-to-earnings (P/E) ratio is a simple concept: The current market price of a firm's common stock is divided by the firm's annual earnings per share. Although the current market price is an unambiguous variable, earnings are not. For example, the earnings variable sometimes represents earnings per share for the most recent year. In that case, the P/E ratio is referred to as the *current P/E ratio* or *trailing P/E ratio*. There is yet another commonly used definition of the P/E ratio that defines earnings per share by using analysts' forecasts of the next year's earnings. This is the *forward P/E ratio*.

Review 1: Relative valuation – trailing (forward) P/E

	Share Price	Current EPS	Current/ Trailing P/E Ratio	Forecast EPS	Forward P/E Ratio
BASF AG (BF)	\$ 70.47	\$ 5.243	13.44	\$ 7.27	9.69
Bayer AG (BAY)	35.64	1.511	23.59	2.69	13.27
Dow Chemical (DOW)	47.40	4.401	10.77	5.71	8.30
E I DuPont (DD)	41.00	2.572	15.94	3.04	13.48
Eastman Chemical (EMN)	51.69	5.75	8.99	5.93	8.71
FMC (FMC)	59.52	5.729	10.39	5.66	10.51
Rohm & Hass (ROH)	45.02	2.678	16.81	3.12	14.44
		Average	14.28		11.20

Problem 1

8-5 VALUING A PRIVATELY HELD FIRM The auto parts business has three large publicly held firms: O'Reilly Automotive Inc. (Orly), Advance Auto Parts Inc. (AAP), and Auto Zone Inc. (AZO). In addition to these publicly held firms, Carquest is the largest privately held firm in the industry. Assume that in summer 2010, your investment banking firm is considering whether to approach the top management of O'Reilly with a proposal that they consider acquiring Carquest, which owns over 4,000 auto parts outlets in the United States. As a preliminary step in the evaluation of the possible acquisition, you have assembled a team of analysts to prepare a preliminary analysis of acquisition price multiples that might be warranted based on current market conditions. The analyst team went to work immediately and compiled the following set of financial information and potential valuation ratios:

Step 1

- a. How would you use this information to evaluate a potential offer to acquire Carquest's equity?
- b. What do you think is driving the rather dramatic differences in the valuation ratios of the three firms?

Problem 1 (cont.)

Financial Information (Millions)	O'Reilly	Advance	Auto Zone
Revenues	\$ 2,120.00	\$ 4,400.00	\$ 5,890.00
EBITDA	321.86	544.38	1,130.00
Net income	171.62	240.16	562.44
Earnings per share	1.507	2.183	7.301
Interest-bearing debt	120.00	560.00	1,720.00
Common equity	1,145.77	939.51	641.16
Total assets	1,713.90	2,615.73	4,401.85
Financial Ratios			
Debt to equity	10.5%	59.6%	269.3%
Gross margins	44.0%	47.3%	49.1%
Operating margins	12.47%	9.42%	16.77%
Expected growth in EPS (5 yrs.)	18.5%	16.0%	13.0%
Market Valuations (Millions)			
Market capitalization	\$ 3,240	\$ 3,040	\$ 6,290
Enterprise value	3,360	3,600	8,010
Valuation Ratios			
Enterprise value/EBITDA	10.44	6.61	7.09
P/E ratio (trailing)	19.42	13.30	11.56
P/E ratio (forward)	15.24	11.21	10.21
Beta	1.24	1.79	1.25

Solvency

Profitability

Growth

Step 2

Problem 1 (cont.)

Answer

- a. The initial estimate would involve using the average valuation ratios for the peer group firms to place a value on Carquest. For example, use Enterprise value/EBITDA and the P/E ratio. Next, we should look at Carquest to see which of the peer firms is most similar in terms of size, growth, and use of financial leverage. This would help us shade our initial analysis toward the most similar firm.

Problem 1 (cont.)

- b. O' Reilly seems to be the "hottest" company. EBITDA as well as PE multiples are the highest. There are significant differences in leverage across the firms. Financial leverage (like operating leverage). O'Reilly has the least financial risk (see Debt to equity ratio). Its profit margin is also pretty good. Growth rate in EPS is also the highest. Lower risk and higher growth justify high multiples for O'Reilly. Although Advance has higher growth and lower financial risk than AutoZone, its operating margins are the lowest. Also beta for Advance is the highest. High beta might arise due to a higher operating leverage. It is not altogether surprising that Autozone has a slightly higher multiple than Advance.

Review 2: P/E discussion

$$\frac{\text{Terminal Value}_{2020}}{\text{Terminal Value}_{2020}} = \frac{\text{FCF}_{2020} \left(1 + \frac{\text{Terminal Growth Rate}(g)}{\text{Cost of Capital } (k_{WACC})} \right)}{\left(\text{Cost of Capital } (k_{WACC}) - \frac{\text{Terminal Growth Rate}(g)}{\text{Terminal Value}_{2020}} \right)} \quad (9.2)$$

P/E Multiples for Stable-Growth Firms

A stable-growth firm is one that is expected to grow indefinitely at a constant rate. The P/E multiple of such a firm is determined by its constant rate of growth and can be calculated by solving the Gordon growth model applied to the valuation of a firm's equity, as in Equation (8.9):⁹

$$P_0 = \frac{\text{Dividend per Share}_0(1 + g)}{k - g} = \frac{\text{Earnings per Share}_0(1 - b)(1 + g)}{k - g} \quad (8.9)$$

where

b = the retention ratio, or the fraction of firm earnings that the firm retains, implying that $(1 - b)$ is the fraction of firm earnings paid in dividends

g = the growth rate of these dividends

k is the required rate of return on the firm's equity

By rearranging the terms in this equation, we can express the P/E ratio as follows:

$$\text{P/E Ratio} = \frac{(1 - b)(1 + g)}{k - g} \quad (8.10a)$$

Review 2: P/E discussion

projects. We assume that these positive NPV investments earn a rate of return, which we denote as r , that exceeds the firm's required rate of return, k . Well-positioned firms with competitive advantages, intellectual property, patents, and managerial expertise can generate both higher rates of return on new investment as well as opportunities to reinvest more of their earnings. It is the combination of the amount by which r exceeds k and the fraction of firm earnings that can be profitably reinvested each year $(1 - b)$ that determines the firm's P/E ratio.

To illustrate, suppose that a firm with \$100 million of invested capital generates returns on invested capital of 20% per year. If the firm pays out 100% of its earnings each year, then the dividend will be \$20 million each year. In this case, the growth rate of the firm's dividends, g , is zero. Now suppose that the firm pays only 40% of its earnings in dividends and reinvests the remaining 60% in its operations. In this case, the first dividend will be $D_1 = \$100\text{M} \times (.2) \times (.4) = \8 million, leaving the firm with a capital base of \$112 million. If the firm continues to generate the same 20% return on the larger capital base, and if it continues to pay out 40% of its earnings and retain 60%, its second-year dividend will be $D_2 = \$112\text{M} \times (.2) \times (.4) = \8.96 million. Dividends will continue to grow at $(\$8.96 - \$8.00)/\$8.00 = 12\%$ annually as long as the dividend payout policy continues and the firm's return on capital remains equal to 20%.

Under these assumptions, we can express a firm's dividend growth rate as the product of its retention rate, b , and the rate of return it can provide on newly invested capital, r . We can now restate the P/E multiple from Equation (8.10a) as follows:

Growth rate (g)
= Retention Ratio (b) $\times$ ROE(r)

$$\text{P/E Multiple} = \frac{(1 - b)(1 + g)}{k - g} = \frac{(1 - b)(1 + br)}{k - br} \quad (8.10b)$$

Review 2: P/E discussion

Table 8-5 The P/E Multiple and Dividend Policy for a Stable-Growth Firm

Firm A: $r = 10\%$, $k = 8\%$			Firm B: $r = 10\%$, $k = 12\%$		
Retention Ratio (b)	Growth Rate (g)	P/E Multiple	Retention Ratio (b)	Growth Rate (g)	P/E Multiple
0%	0%	12.50	0%	0%	8.33
10%	1%	12.99	10%	1%	8.26
20%	2%	13.60	20%	2%	8.16
30%	3%	14.42	30%	3%	8.01
40%	4%	15.60	40%	4%	7.80
50%	5%	17.50	50%	5%	7.50
60%	6%	21.20	60%	6%	7.07
70%	7%	32.10	70%	7%	6.42
80%	8%	Undefined*	80%	8%	5.40
90%	9%	Undefined*	90%	9%	3.63
100%	10%	Undefined*	100%	10%	0.00

r = return on reinvested earnings

k = stockholders' required rate of return

*These cells have negative P/E multiples. Here the formula for P/E multiple $(1 - b)(1 + g)/(k - g)$ does not apply because, for $k < g$, the series (perpetuity) does not converge. Intuitively, this situation implies that the earnings are growing faster than the cost of capital forever. This situation cannot arise in a competitive economy, where free entry will drive down high growth rates to the feasible range (where $k < g$).

Review 2: P/E discussion

P/E Multiple for a High-Growth Firm

We can relate the P/E multiple for a high-growth firm to company fundamentals in much the same way as we did for a stable-growth firm—with one important difference. Because we do not expect a firm to be able to achieve high growth forever, describing the firm's growth prospects requires two growth periods. We assume that the firm experiences very high growth lasting for a period of n years, followed by a period of much lower but stable growth.¹⁰ Equation (8.11) captures the valuation of the equity of such a firm:

$$P_0 = \frac{\text{EPS}_0(1 + g_1)(1 - b_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right) + \frac{\text{EPS}_0(1 + g_1)^n(1 - b_2)(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right) \quad (8.11)$$

¹⁰ Recall that we are describing a two-stage growth model similar to our discussion in Chapter 4. The difference here is that we define equity value as the present value of expected future dividends (not equity free cash flow).

Review 2: P/E discussion

First stage (Planning period)

- The dividend at year t : $Div_t = EPS_0(1 + g_1)^t(1 - b)$
- The total PV of dividends at year 0: $\sum \frac{EPS_0(1+g_1)^t(1-b)}{(1+k)^t}$
- Factor out the constants: $EPS_0(1 - b) \sum \frac{(1+g_1)^t}{(1+k)^t}$
- *first term* = $\frac{1+g_1}{1+k}$; *common ratio* = $\frac{1+g_1}{1+k}$
- $EPS_0(1 - b) \times \frac{\frac{1+g_1}{1+k} \times (1 - (\frac{1+g_1}{1+k})^n)}{1 - \frac{1+g_1}{1+k}}$

$$\frac{EPS_0(1 + g_1)(1 - b_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right)$$

Geometric Series Formula

$$S_n = a_1 \left(\frac{1 - r^n}{1 - r} \right)$$

WHERE:

a_1 = first term

r = common ratio

n = number of terms

Review 2: P/E discussion

Second stage (Terminal value)

- Borrowing from Gordon growth model

$$P_0 = \frac{\text{Dividend per Share}_0(1 + g)}{k - g} = \frac{\text{Earnings per Share}_0(1 - b)(1 + g)}{k - g} \quad (8.9)$$



EPS in year n

Discounted to present

$$\frac{\text{EPS}_0(1 + g_1)^n(1 - b_2)(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right)$$

Review 2: P/E discussion

In this equation we abbreviate earnings per share₀ as EPS₀, and both the retention ratio (b) and the growth rate (g) are now subscripted to reflect the fact that the firm's dividend policy and its growth prospects can be different for the two growth periods.¹¹

We can interpret Equation (8.11) as follows: The first term on the right-hand side expresses the present value of a stream of dividends received over the next n years. For the high-growth firm, dividends grow at a rapid rate equal to g_1 for n years, after which the dividends grow at a constant (and lower) rate g_2 forever.

We can now solve for the P/E multiple simply by dividing both sides of Equation (8.11) by EPS₀. The result is found below:

$$\begin{aligned} \frac{P_0}{\text{EPS}_0} = & \frac{(1 - b_1)(1 + g_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right) \\ & + \frac{(1 - b_2)(1 + g_1)^n(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right) \end{aligned} \quad (8.12)$$

Problem 2

8-9 HIGH-GROWTH OR STABLE-GROWTH FIRM? Intel Corporation is a leading manufacturer of semiconductor chips. The firm was incorporated in 1968 in Santa Clara, California, and represents one of the greatest success stories of the computer age. Although Intel continues to grow, the industry in which it operates has matured, so there is some question whether the firm should be evaluated as a high-growth company or stable-growth company from now on. For example, in December 2007, the firm's shares were trading for \$20.88, and has a price-earnings ratio of 17.622. Compared to Google Inc.'s price-earnings ratio of 53.71 on the same date, it would appear that the decision has already been made by the market.

Intel's expected earnings for 2007 were \$1.13 per share, and its payout ratio was 48%. Furthermore, selected financial data for the sector, industry, and seven of the largest firms (including Intel) are found in Exhibit P8-9.1 on page 301.

- a. Is Intel's current stock price of \$20.88 reasonable in light of its sector, industry, and comparison firms?

Problem 2

- b. Intel has a beta coefficient equal to 1.66. If we assume a risk-free rate of 5.02% and a market risk premium of 5%, what is your estimate of the required rate of return for Intel's stock using CAPM? What rate of growth in earnings is consistent with Intel's policy of paying out 48% of earnings in dividends and the firm's historical return on equity? Using your estimated growth rate, what is the value of Intel's shares using the Gordon (single-stage) growth model? Analyze the reasonableness of your estimated value per share using the Gordon model.
- c. Using your analysis in Problem 8-9(b), what growth rate is consistent with Intel's current share price of \$20.88?
- d. Analysts expect Intel's earnings to grow at a rate of 12% per year over the next five years. What rate of growth from year 6 forward (forever) is needed to warrant Intel's current stock price (use your CAPM estimate of the required rate of return on equity)? (*Hint: Use a two-stage growth model where Intel's earnings grow for five years at 12% and from year 6 forward at a constant rate.*)

Problem 2

Exhibit P8-9.1 Industry Comparables for Intel Corporation

Description	Market Cap	P/E	Return on Equity %	Dividend Yield %	Long-Term Debt to Equity	Price to Book Value	Net Profit Margin	Price to Free Cash Flow
Sector: technology	5,344.81B	27.716	14.77%	1.90%	0.691	5.588	10.39%	55.435
Industry: semiconductor — broad line	252.89B	19.9	16.20%	1.30%	0.096	3.42	15.50%	193.3
Intel Corp.	120.51B	17.622	19.63%	1.90%	0.064	3.437	18.72%	121.039
Texas Instruments Inc.	44.62B	11.08	22.94%	0.50%	0.004	3.71	18.67%	−5,577.55
STMicroelectronics NV	16.35B	24.959	7.81%	0.70%	0.209	1.764	8.24%	−11.219
Advanced Micro Devices Inc.	11.79B	21.152	12.61%	0.00%	0.138	2.088	10.13%	−58.916
Analog Devices Inc.	11.48B	22.667	15.42%	1.90%	NA	3.342	21.48%	311.392
Maxim Integrated Products Inc.	10.28B	23.025	16.93%	1.90%	NA	3.681	21.39%	NA
National Semiconductor Corp.	8.04B	18.49	25.67%	0.60%	0.012	4.481	22.18%	154.483

Problem 2

- a. Yes, Intel's current stock price of \$20.88 appears **reasonable**. Although its P/E ratio is slightly lower than the industry average, this is justified by Intel's strong profitability (high ROE), low financial risk (low debt-to-equity), and consistent dividend payouts. Compared to peers, Intel is positioned as a relatively stable, mature company rather than a high-growth one, which supports a slightly lower P/E multiple.

Problem 2

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$b. \quad k_e = 5.02 + 1.66 \times 5\% = 13.32\%$$

$$\text{P/E Multiple} = \frac{(1 - b)(1 + g)}{k - g} = \frac{(1 - b)(1 + br)}{k - br} \quad (8.10b)$$

$$P = \frac{EPS_1(1 - b)}{k - br}$$

$$\bullet \quad r = (1 - 48\%) \times 19.63\%$$

$$\bullet \quad P = \frac{1.13 \times 48\%}{13.32\% - (1 - 48\%) \times 19.63\%} = 17.43$$

$$c. \quad 20.88 = \frac{1.13 \times 48\%}{13.32\% - g}, g = 10.72\%$$

Problem 2

$$P_0 = \frac{\text{EPS}_0(1 + g_1)(1 - b_1)}{k - g_1} \left(1 - \frac{(1 + g_1)^n}{(1 + k)^n} \right) + \frac{\text{EPS}_0(1 + g_1)^n(1 - b_2)(1 + g_2)}{k - g_2} \left(\frac{1}{(1 + k)^n} \right) \quad (8.11)$$

$$d. \quad \$20.88 = \frac{1.13 \times 48\%}{13.32\% - 12\%} \times \left(1 - \frac{(1 + 12\%)^5}{(1 + 13.32\%)^5} \right) + \frac{1.13 \times (1 + 12\%)^4 \times 48\% \times (1 + g_2)}{13.32\% - g_2} \times \left(\frac{1}{(1 + 13.32\%)^5} \right), \quad g = 10.60\%$$

- Hint: g is found using Goal Seek in Excel.

Review 3: The Pricing of Initial Public Offering (IPO)

Step 1: Initial pricing range determination

- The lead underwriter determines an initial estimate of the range of values for the issuer's shares (e.g., \$18 - \$22).

Step 2: Road show and bookbuilding

- After setting the initial price range, the underwriters and company executives conduct a roadshow, travelling to meet potential institutional investors.

Step 3: Final offering price decision

- On the eve of the IPO, the underwriters and company executives meet to decide on the final offering price (e.g., \$19).

Problem 3

PROBLEM 8-12 **MINI-CASE** VALUING GOOGLE IPO SHARE PRICE¹⁶

Google Inc. of Mountain View, California, operates the most popular and powerful search engine on the Web. The company went public using an unconventional Dutch auction method on August 19, 2004. The resulting IPO was the largest

Internet IPO ever, raising \$1.67 billion and leaving the firm with 271,219,643 shares of common stock.

While Google commands a wide lead over its competitors in the search engine market, it is witnessing increased pressure from well-funded rival entities. Yahoo! Inc., with a market cap of approximately \$38.43 billion, is generally regarded as following a business model very similar to Google's.

- a. Use the data found in Exhibit P8-12.1 for the following companies as comparables in your analysis: Earthlink, Yahoo!, eBay, and Microsoft. Compute the IPO value of Google: shares using each of the comparable firms separately, and then use an average "multiple" of the comparable firms. Use the year-end 2003 balance sheets and income statements of the comparable firms to do the analysis. Assume that Google's forecasted values at the time of the IPO are as follows: Net income is \$400 million, EBITDA is approximately \$800 million, cash and equivalents are \$430 million, and interest-bearing debt (total short-term and long-term) equals only \$10 million.¹⁷
- b. Which of the four comparable firms do you think is the best comparison firm for Google? Why?
- c. How has the stock performed after the IPO? Do you believe that Google is currently correctly valued in the stock market? Explain your answer.

Problem 3

Exhibit P8-12.1

Financial Information	Earthlink (ELINK)	Yahoo! (YHOO)	eBay (EBAY)	Microsoft (MSFT)
2003 shares outstanding	159,399,000	655,602,000	646,819,000	10,800,000,000
2003 fiscal close stock price	\$ 10.00	\$ 45.03	\$ 64.61	\$ 25.64
Market capitalization	\$1,593,990,000	\$29,521,758,060	\$41,790,975,590	\$276,912,000,000
Short-term debt	\$ 900,000	\$ 0	\$ 2,800,000	\$ 0
Long-term debt	\$ 0	\$ 750,000,000	\$ 124,500,000	\$ 0
Cash and equivalents	\$ 349,740,000	\$ 713,539,000	\$ 1,381,513,000	\$ 6,438,000,000
Short-term investments	\$ 89,088,000	\$ 595,975,000	\$ 340,576,000	\$ 42,610,000,000
EBITDA	\$ 218,100,000	\$ 455,300,000	\$ 818,200,000	\$ 14,656,000,000
Net income	\$ (62,200,000)	\$ 237,900,000	\$ 441,800,000	\$ 9,993,000,000
Calculated EPS	(0.39)	0.36	0.68	0.93

Problem 3

$$\begin{aligned}\text{Owner's Equity} &= \text{Enterprise Value} - (\text{Interest-Bearing Debt} - \text{Cash}) \quad (8.5a) \\ &= \text{Enterprise Value} - \text{Net Debt}\end{aligned}$$

$$\frac{\text{Enterprise Value}_{\text{Airgas 2014}}}{\text{EBITDA}_{\text{Airgas 2014}}} = \text{EBITDA Multiple} \quad (8.6)$$

a. ELINK - EBITDA Multiple

- *Enterprise Value* = 1,593,990,000 + 900,000 – 349,740,000 = 1,245,150,000
- *EBITDA Multiple* = $\frac{1,245,150,000}{218,100,000} = 5.71$
- *Enterprise Value (GOOGL)* = 5.71 × 800,000,000 = 4,568,000,000

Problem 3

- *Owner's equity (GOOGL)* = $4,568,000,000 - (10,000,000 - 430,000,000) = 4,988,000,000$
- *IPO Price (GOOGL)* = $\frac{4,988,000,000}{271,219,643} = 18.39$

ELINK – P/E Ratio

- *Price – to – earnings* = $\frac{10}{(0.39)} = (25.64)$
- *IPO Price* = $(25.64) \times \frac{400,000,000}{271,219,643} = (37.69)$

	Earthlink	Yahoo	eBay	Microsoft	Average
Imputed IPO price per share from PE ratio	(\$38)	\$183.01	\$139.51	\$40.87	\$81.40
Impute EV from EBITDA multiples	\$4,567,262,724	\$51,936,251,368	\$39,635,064,864	\$14,763,864,629	\$27,725,610,896
Owner's equity	\$4,987,262,724	\$52,356,251,368	\$40,055,064,864	\$15,183,864,629	\$28,145,610,896
Impute IPO price per share	\$18.39	\$193.04	\$147.68	\$55.98	\$103.77

Problem 3

- b.** **Yahoo** seems to be the closest in terms of leverage, and it is. And fairly close in terms of EBITDA, NI. Microsoft and Earthlink have little debt. Microsoft seems to be much bigger in size. e-Bay has huge amounts of debt. It does not seem comparable. The choice is between Earthlink and Yahoo. Yahoo may be the best. Its debt is comparable to Google, its EBITDA and NI are closer to Google, and it has a similar businesses.

Problem 3

NasdaqGS - Nasdaq Real Time Price • USD

Alphabet Inc. (GOOG)

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163.85 +2.38 +(1.47%) 164.50 +0.65 +(0.40%)

At close: April 25 at 4:00:02 PM EDT

Pre-Market: 7:38:04 AM EDT



Previous Close	161.47	Day's Range	163.03 - 168.24	Market Cap (intraday)	1.954T	Earnings D...	Jul 21, 2025 - Jul 25, 2025
Open	167.10	52 Week Range	142.66 - 208.70	Beta (5Y Monthly)	1.03	Forward Dividend & Yield	0.80 (0.49%)
Bid	163.50 x 100	Volume	35,148,053	PE Ratio (TTM)	18.29	Ex-Dividend Date	Mar 10, 2025
Ask	168.00 x 300	Avg. Volume	22,904,529	EPS (TTM)	8.96	1y Target Est	201.73

Problem 3

- c. Since its IPO, Google (now Alphabet Inc.) has delivered **extraordinary stock performance (65X)**. Alphabet's valuation today is fair because it reflects a balance of strong earnings, moderate but reliable growth, low financial risk, and reasonable market expectations — without extreme optimism or pessimism built into the price.

- *Thank you for your time and attention!*